

- 4 The photograph shows some leaves of *Piper betle*, a plant that grows in the Philippines and other countries in southeast Asia.



(Source: © Bowonpat Sakaew/Alamy Stock Photo)

The leaves are eaten as a traditional cure for human digestive disorders.

The leaves are thought to prevent the growth of some species of bacteria.

A student formed the following hypothesis:

The leaves of Piper betle contain antibacterial compounds that reduce bacterial growth.

Plan an investigation to find evidence to support or reject this hypothesis.

Your answer should give details under the following headings.

- (a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

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(b) Devise a detailed method, including how you would control and monitor important variables.

(9)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.

(d) Describe **two** limitations of your proposed method.

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(Total for Question 4 = 16 marks)

TOTAL FOR PAPER = 50 MARKS